Lecture 7: Fourier Transform

CSE 219: Signals and Linear Systems

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Fourier Transform: to ∞ and beyond

Looking inside any function. Yes, not just periodic ones!

Aperiodic functions crying in the background...



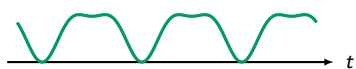
Fourier Transform: When the period goes to infinity!

Fourier series decomposes **periodic** signals.
Fourier transform lets us do something similar for the **aperiodic** ones.

Fourier series had one big condition

Fourier Series

$$x(t+T) = x(t)$$



keeps repeating

But real signals often do not repeat

no fixed period



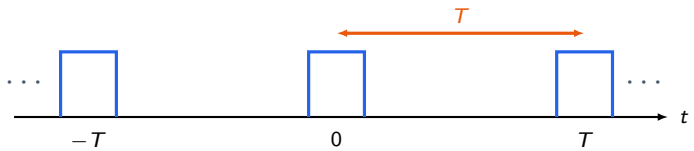
one-time event

Can we still describe a non-repeating signal using frequencies?

Let's cheat first: repeat one lonely pulse

Start with one finite-duration signal $x(t)$ and repeat it!

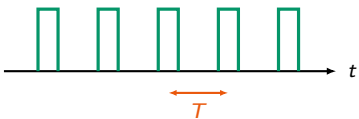
$$\widetilde{x}_T(t) = \sum_{m=-\infty}^{\infty} x(t - mT)$$



Now it is periodic, so Fourier series works.

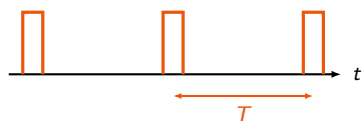
Now guess what? We'll just keep stretching the period ...

Small T



copies are close

Large T



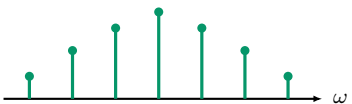
copies move away

As $T \to \infty$, the repeated signal starts looking like one isolated signal.

And what happens in frequency?

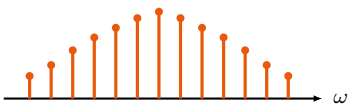
$$\omega_0 = \frac{2\pi}{T}, \qquad \Delta\omega = \omega_0.$$

Small T, large $\Delta\omega$



few frequency lines

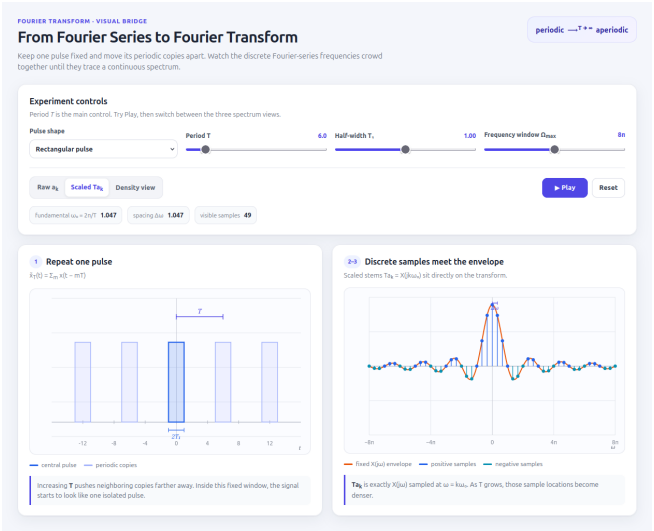
Large T, small $\Delta\omega$



many frequency lines

Increasing T makes the allowed frequencies closer and closer.

Please try this visualization on Fourier Transforms!



Fourier-series coefficients of the repeated signal

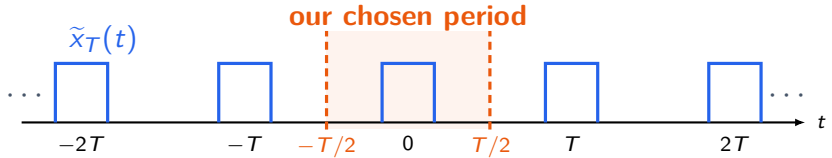
The periodic signal has Fourier series

$$\tilde{x}_T(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

$$a_k = \frac{1}{T} \int_{\text{one period}} \tilde{x}_T(t) e^{-jk\omega_0 t}\, dt$$

This is still the same Fourier-series coefficient formula.

Let's choose one period that contains the pulse



$$a_k = \frac{1}{T}\int_{-T/2}^{T/2} \widetilde{x}_T(t)e^{-jk\omega_0 t}\,dt$$

Inside this period, $\widetilde{x}_T(t) = x(t)$, the isolated pulse.

$$\text{Thus, } a_k = \frac{1}{T}\int_{-\infty}^{\infty} x(t)e^{-jk\omega_0 t}\,dt$$

A smooth spectrum is hiding behind the stems

Look at the integral part

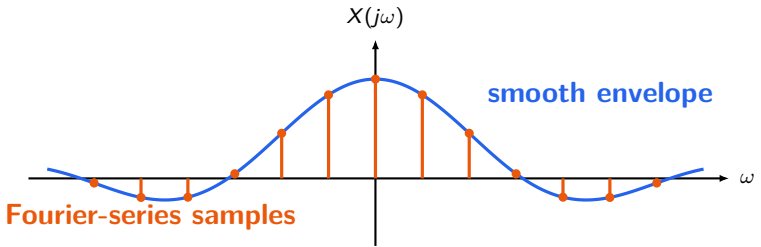
$$a_k = \frac{1}{T} \underbrace{\int_{-\infty}^{\infty} x(t) e^{-jk\omega_0 t}\, dt}_{\text{depends on } k\omega_0}$$

Let

$$\boxed{X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t}\, dt}$$

So the stems are just samples of a smooth curve!

$$Ta_k = X(jk\omega_0)$$



As T increases, the sample locations $k\omega_0$ become closer.

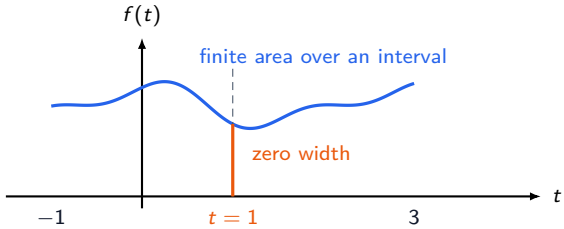
A Detour

How do we convert something discrete into something continuous?

From *discrete* to *continuous*...



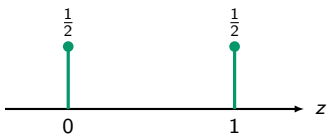
Can one single point contain any area?



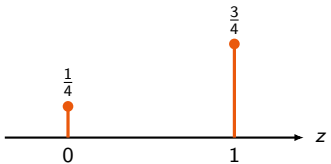
The area of a single vertical line is **zero**,
even though a whole interval can contain plenty of area.

Let's take coins for example . . .

Fair coin



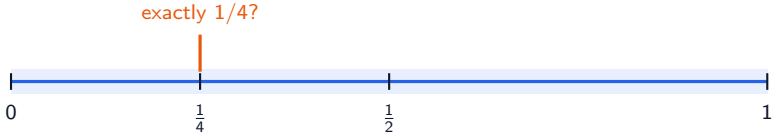
Biased coin



Only two outcomes. Giving each point its own probability is easy!

But what if the answer can be anywhere?

Choose a real number uniformly at random from $[0, 1]$.

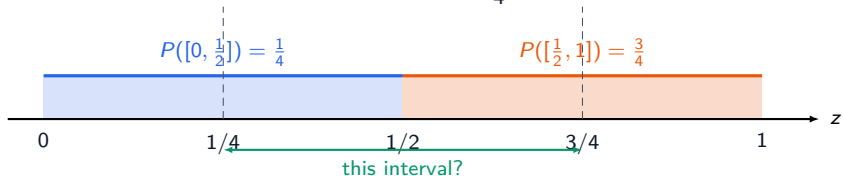


$P(Z = \frac{1}{4}) = 0$ even though $\frac{1}{4}$ is a perfectly possible outcome.

But $P(\frac{1}{4} \leq Z \leq \frac{1}{2}) = \frac{1}{4}$ — an **interval** has nonzero probability.

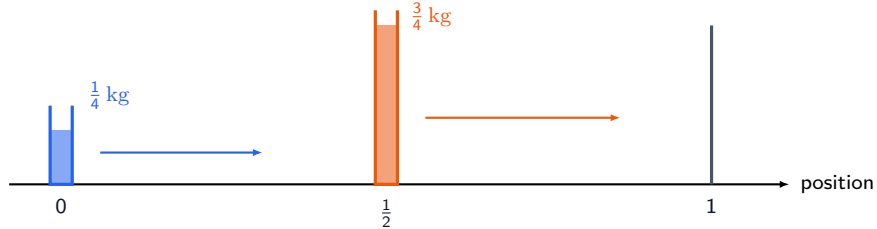
A continuous choice can be biased too

Suppose the first half gets probability $\frac{1}{4}$, and the second half gets $\frac{3}{4}$.



If the choice is uniform *within each half*, how much probability lies in $[\frac{1}{4}, \frac{3}{4}]$?

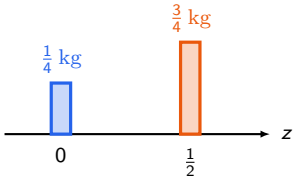
Let's pretend probability is water!



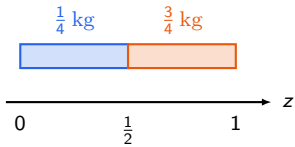
What if the first tube spreads its water across $[0, \frac{1}{2}]$,
and the second spreads across $[\frac{1}{2}, 1]$?

What if we spread the water out?

Before: water in two tubes



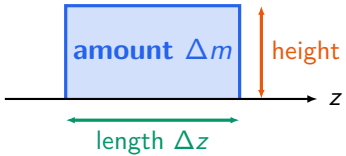
After: water spread out



We know how much water each interval contains.

But what should the **height** of our new graph represent?

Let's give the spread-out water a height!

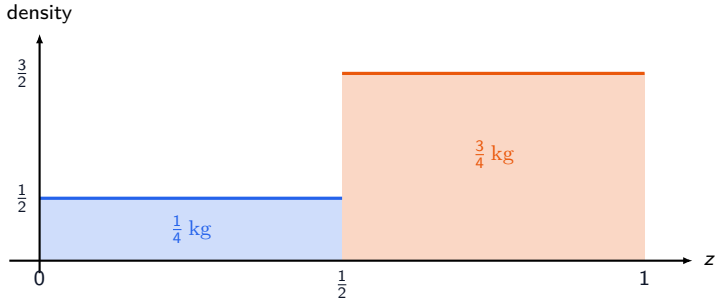


$$\text{Amount} = \text{Height} \times \text{Length}$$

Let's call this height the **density**!

$$\boxed{\text{Density} = \frac{\text{Amount}}{\text{Length}}}$$

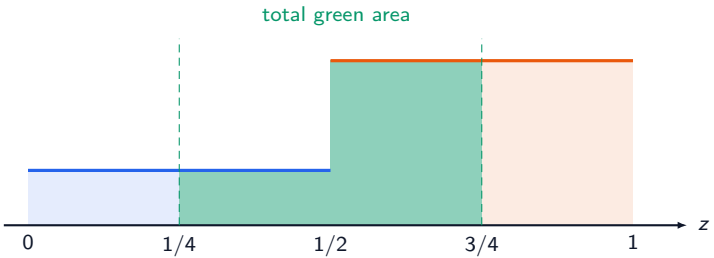
The water spreads, but the amount stays the same



$$\frac{\text{amount}}{\text{length}} = \frac{1/4}{1/2} = \frac{1}{2}$$

$$\frac{\text{amount}}{\text{length}} = \frac{3/4}{1/2} = \frac{3}{2}$$

Now let's use density to find what we want!



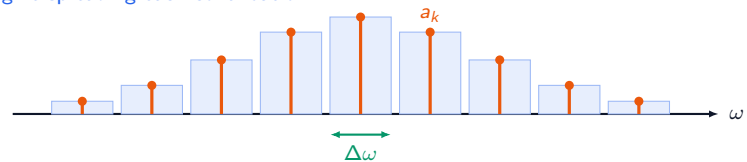
$$P\left(\frac{1}{4} \le Z \le \frac{3}{4}\right) = \underbrace{\frac{1}{4} \cdot \frac{1}{2}}_{1/8} + \underbrace{\frac{1}{4} \cdot \frac{3}{2}}_{3/8} = \boxed{\frac{1}{2}}$$

Amount in a range = area under its density curve.

And now, back to our frequency stems!

Each Fourier-series coefficient a_k belongs to one allowed frequency $k\omega_0$.

imagine spreading each contribution



The analogy is about **contribution per frequency interval**.
(Unlike water, Fourier coefficients can also be negative or complex.)

So what is the density of a Fourier coefficient?

From our water example, a natural choice is

$$\text{density} = \frac{\text{amount}}{\text{interval width}} \qquad \longrightarrow \qquad \frac{a_k}{\Delta\omega}.$$

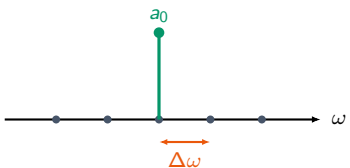
With angular frequency, we use the scaled density

$$\boxed{\frac{2\pi a_k}{\Delta\omega}}$$

so the inverse transform carries the familiar $1/(2\pi)$.

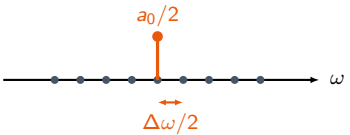
But wait... what happens when we double the period?

Period T



$$a_0 = \frac{2T_1}{T}$$

Period $2T$



$$a_0' = \frac{2T_1}{2T} = \frac{a_0}{2}$$

The coefficient becomes half and so does the frequency spacing!

The coefficient shrinks, but the density survives!

$$\text{Frequency density} = \frac{2\pi a_k}{\Delta\omega}$$

Period T

$$\frac{2\pi a_0}{\Delta\omega}$$

Period $2T$

$$\frac{2\pi(a_0/2)}{\Delta\omega/2} = \frac{2\pi a_0}{\Delta\omega}$$

Just like water: when the amount and the interval width shrink together, the density remains unchanged!

Let's plug in the frequency spacing as we know. . . !

Recall that, $\Delta\omega = \omega_0 = \dfrac{2\pi}{T} \qquad \Longrightarrow \qquad \dfrac{2\pi}{\Delta\omega} = T.$

So, our frequency density $\dfrac{2\pi a_k}{\Delta\omega} = Ta_k.$

But we already know, $a_k = \dfrac{1}{T}\displaystyle\int_{-\infty}^{\infty} x(t)e^{-jk\omega_0 t}\,dt.$

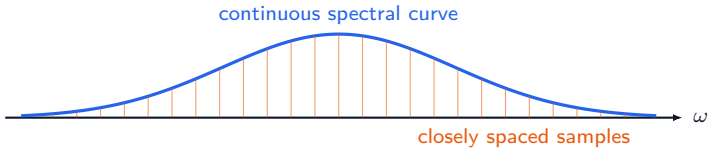
Now let's arrive at a simple beautiful expression!

$$\text{Therefore, } \frac{2\pi a_k}{\Delta\omega} = Ta_k = \int_{-\infty}^{\infty} x(t)e^{-jk\omega_0 t}\,dt$$

Interesting! The expression depends on the frequency $k\omega_0$.
But what if ω could take **any real value**?

Let's give the continuous spectrum a name!

The sampled frequencies $k\omega_0$ get arbitrarily close as $T \to \infty$.



$$X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}\,dt$$

This is the **Fourier transform analysis equation**.

But breaking down into ingredients was only half the story...

Recall the Fourier-series synthesis equation, the return ticket:

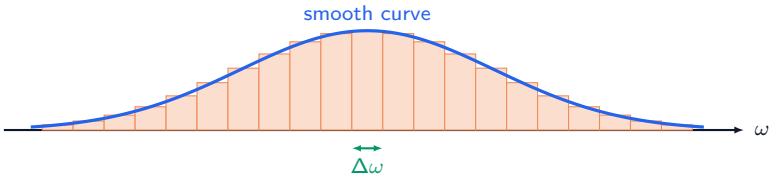
$$\widetilde{x}_T(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}.$$

But we just found

$$a_k = \frac{\Delta\omega}{2\pi} X(jk\omega_0).$$

$$\boxed{\widetilde{x}_T(t) = \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} X(jk\omega_0) e^{jk\omega_0 t}\, \Delta\omega}$$

Tiny frequency strips become an integral



$T \rightarrow \infty \Rightarrow \Delta\omega \rightarrow 0$: a sum of narrow frequency strips becomes an integral.

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} \, d\omega$$

Our two-way ticket: the Fourier transform pair

Analysis: time $\longrightarrow$ frequency

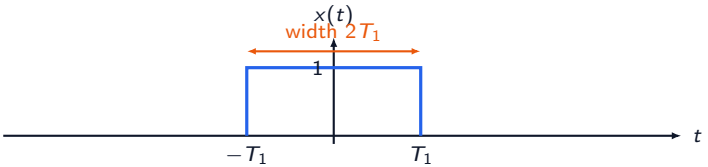
$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t}\, dt$$

Synthesis: frequency $\longrightarrow$ time

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t}\, d\omega$$

Our first transform: one lonely rectangle

$$x(t) = \begin{cases} 1, & |t| < T_1, \\ 0, & \text{otherwise.} \end{cases}$$

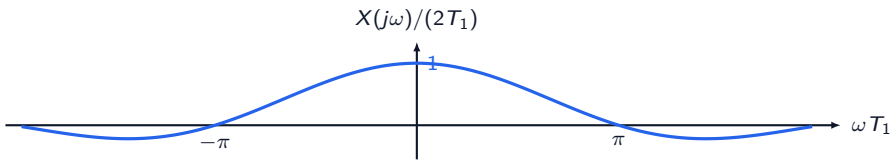


The signal is zero almost everywhere.
Why integrate where there is nothing?

The rectangle turns into a sinc-shaped spectrum

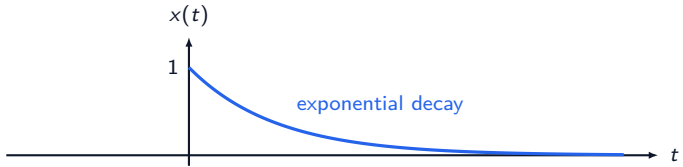
$$X(j\omega) = \int_{-T_1}^{T_1} e^{-j\omega t}\, dt = \left[\frac{e^{-j\omega t}}{-j\omega}\right]_{-T_1}^{T_1}$$

$$\boxed{X(j\omega) = \frac{2\sin(\omega T_1)}{\omega}} \qquad X(0) = 2T_1$$



One more classic: a signal that fades away

$$x(t) = e^{-t}u(t)$$



The signal begins at $t = 0$. So the transform integral begins there too.

A fading signal has a beautifully simple transform

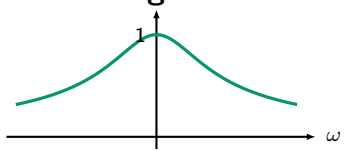
$$
\begin{aligned}
X(j\omega) &= \int_0^\infty e^{-t} e^{-j\omega t}\, dt \\
&= \int_0^\infty e^{-(1+j\omega)t}\, dt \\
&= \left[-\frac{e^{-(1+j\omega)t}}{1+j\omega} \right]_0^\infty .
\end{aligned}
$$

$$
\boxed{e^{-t}u(t) \quad \longleftrightarrow \quad \frac{1}{1+j\omega}}
$$

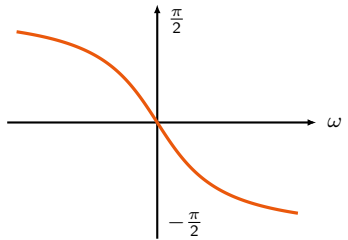
What does the fading signal look like in frequency?

$$X(j\omega) = \frac{1}{1 + j\omega}$$

Magnitude



Phase



$$|X(j\omega)| = \frac{1}{\sqrt{1+\omega^2}}, \qquad \angle X(j\omega) = -\tan^{-1}(\omega).$$

Properties of Fourier Transforms

Pretty similar to the ones of Fourier Series, really!

Same beautiful properties, but with infinitely many frequencies!

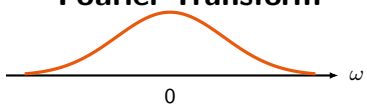
We already learned how operations affect the little waves inside a signal.

Fourier Series



Discrete frequencies $k\omega_0$

Fourier Transform

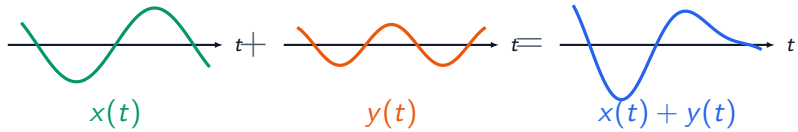


A continuous spectrum in ω

The rules will feel familiar: replace $k\omega_0$ with ω.

Add in time? Just add in frequency!

Linearity survives the move to continuous frequencies.

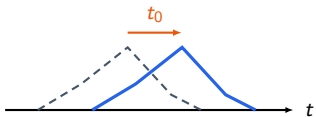


$$\alpha x(t) + \beta y(t) \quad \longleftrightarrow \quad \alpha X(j\omega) + \beta Y(j\omega)$$

Already know both transforms? No more integration needed.

Shift it or flip it, the spectrum remembers!

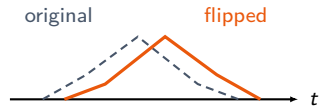
Time shifting



$$x(t - t_0) \longleftrightarrow e^{-j\omega t_0} X(j\omega)$$

Delay changes phase, not magnitude.

Time reversal



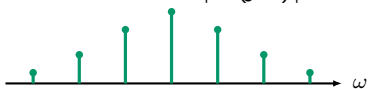
$$x(-t) \longleftrightarrow X(-j\omega)$$

The frequency axis flips as well.

Differentiate a signal, the fast waves react more!

Recall: differentiating $e^{j\omega t}$ produces $j\omega e^{j\omega t}$.

Before: $|X(j\omega)|$



After: $|j\omega X(j\omega)|$



$$\frac{d}{dt}x(t) \quad \longleftrightarrow \quad j\omega X(j\omega)$$

A practice problem

You are given

$$x(t) \longleftrightarrow X(j\omega), \qquad y(t) \longleftrightarrow Y(j\omega).$$

Find the Fourier transform $Z(j\omega)$ of

$$\boxed{z(t) = \frac{d}{dt}\Big[2x(3-t) + y(t-2)\Big].}$$

shift and reverse → scale and add → differentiate

No integration required—only the properties we just recalled.

Answer: unwrap the operations one at a time

First: $x(3-t)$

Shift first:

$$x(t+3) \longleftrightarrow e^{j3\omega} X(j\omega)$$

Then reverse:

$$x(3-t) \longleftrightarrow e^{-j3\omega} X(-j\omega)$$

Second: $y(t-2)$

A delay of 2 gives

$$y(t-2) \longleftrightarrow e^{-j2\omega} Y(j\omega)$$

Now add the two results.

Differentiate *last*: multiply the entire result by $j\omega$.

$$\boxed{Z(j\omega) = j\omega \left[2e^{-j3\omega} X(-j\omega) + e^{-j2\omega} Y(j\omega) \right]}$$

Examples of Fourier Transforms

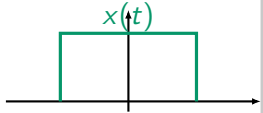
Let's keep rotating and oscillating random things. . . !



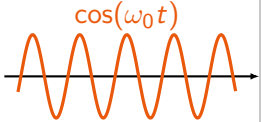
Injecting oscillation into a signal

$$x(t) \quad \longrightarrow \quad x(t)\cos(\omega_0 t)$$

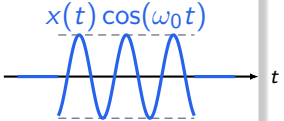
Envelope $x(t)$



Oscillation



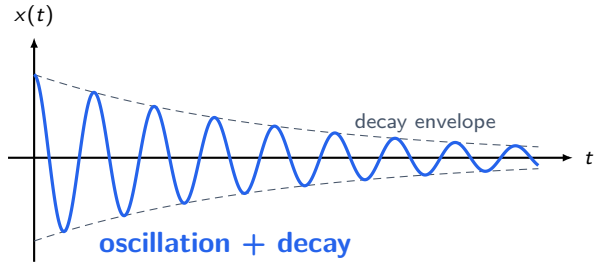
Oscillating envelope



We keep the shape of $x(t)$ as an envelope, but put oscillation inside it.

This happens in real systems too

A damped oscillation



Examples: damped pendulum, vibrating string with loss, ringing circuits, fading echoes.

So what transform should we expect?

$$x(t) \quad \longleftrightarrow \quad X(j\omega).$$

What is the Fourier transform of

$$x(t)\cos(\omega_0 t)?$$

Let's first solve the cleaner version:

$$x(t)e^{j\omega_0 t}.$$

Multiplying by $e^{j\omega_0 t}$ shifts every frequency

A frequency component inside $x(t)$ looks like

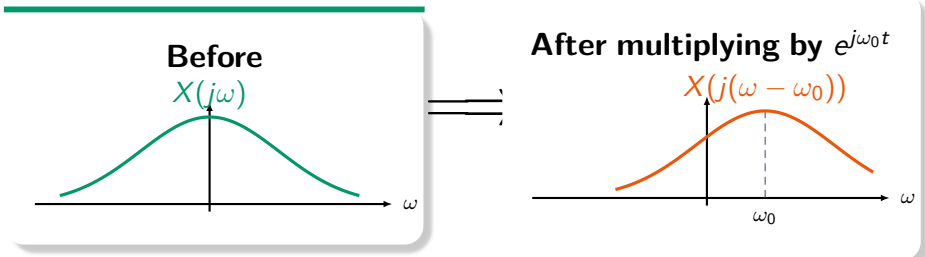
$$e^{j\omega t}.$$

After multiplying by $e^{j\omega_0 t}$,

$$e^{j\omega t} \cdot e^{j\omega_0 t} = e^{j(\omega+\omega_0)t}.$$

Every component at ω moves to $\omega + \omega_0$.

The spectrum slides to the right



To move the spectrum right by ω_0, we write $X(j(\omega - \omega_0))$.

Frequency-shifting property

If

$$x(t) \quad \longleftrightarrow \quad X(j\omega),$$

then

$$\boxed{x(t)e^{j\omega_0 t} \quad \longleftrightarrow \quad X\big(j(\omega-\omega_0)\big)}$$

Multiplication by a complex exponential shifts the spectrum.

Cosine is two rotating exponentials

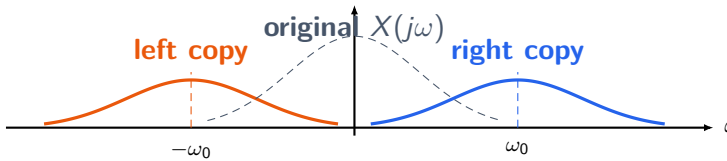
$$\cos(\omega_0 t) = \frac{1}{2}\left(e^{j\omega_0 t} + e^{-j\omega_0 t}\right)$$

So

$$x(t)\cos(\omega_0 t) = \frac{1}{2}x(t)e^{j\omega_0 t} + \frac{1}{2}x(t)e^{-j\omega_0 t}.$$

A cosine creates two shifted copies of the spectrum.

Cosine modulation creates two spectral copies



$$x(t)\cos(\omega_0 t) \quad \longleftrightarrow \quad \frac{1}{2}X\big(j(\omega-\omega_0)\big)+\frac{1}{2}X\big(j(\omega+\omega_0)\big)$$

What did oscillation do?

In time:

$x(t) \to x(t)\cos(\omega_0 t)$

We injected oscillation.

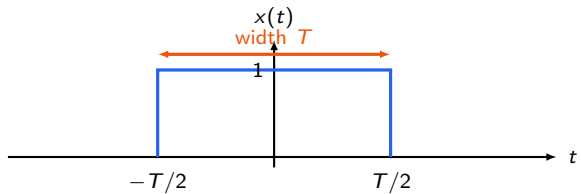
In frequency:

$X(j\omega) \to$ two shifted copies

The spectrum
moves to $\pm\omega_0$.

A very common transform: the rectangle

$$x(t) = \begin{cases} 1, & -\frac{T}{2} < t < \frac{T}{2}, \\ 0, & \text{otherwise.} \end{cases}$$



This transform appears everywhere, let's keep it in our toolbox :D

Fourier transform of the rectangle

Since $x(t) = 1$ only from $-T/2$ to $T/2$,

$$X(j\omega) = \int_{-T/2}^{T/2} e^{-j\omega t}\, dt.$$

$$X(j\omega) = \left[\frac{e^{-j\omega t}}{-j\omega}\right]_{-T/2}^{T/2} = \frac{e^{j\omega T/2} - e^{-j\omega T/2}}{j\omega}.$$

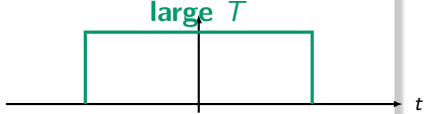
$$\boxed{X(j\omega) = \frac{2\sin(\omega T/2)}{\omega}}$$

Wider rectangle, narrower spectrum

$$x(t) = 1, \quad |t| < \frac{T}{2} \qquad \Longleftrightarrow \qquad X(j\omega) = \frac{2\sin(\omega T/2)}{\omega}.$$

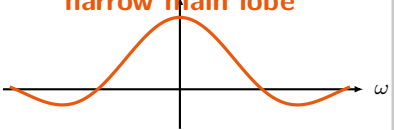
Time domain

large T



Frequency domain

narrow main lobe



Practice: two centered rectangles

$$x_1(t) = \begin{cases} 1, & -1 < t < 1, \\ 0, & \text{otherwise.} \end{cases}$$

$$X_1(j\omega) = ?$$

$$x_2(t) = \begin{cases} 1, & -3 < t < 3, \\ 0, & \text{otherwise.} \end{cases}$$

$$X_2(j\omega) = ?$$

Use

$$\boxed{\text{width } T \implies X(j\omega) = \frac{2\sin(\omega T/2)}{\omega}.}$$

Answer: just plug in the width

For $-1 < t < 1$, the width is $T = 2$:

$$\boxed{X_1(j\omega) = \frac{2\sin(\omega)}{\omega}.}$$

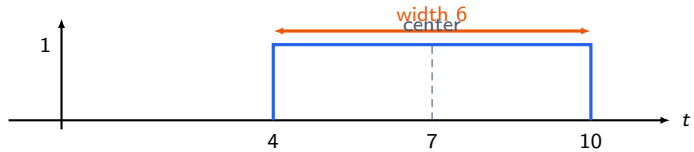
For $-3 < t < 3$, the width is $T = 6$:

$$\boxed{X_2(j\omega) = \frac{2\sin(3\omega)}{\omega}.}$$

The same rectangle formula works once we identify the width.

What if the rectangle is shifted?

$$x(t) = \begin{cases} 1, & 4 < t < 10, \\ 0, & \text{otherwise.} \end{cases}$$



This is a width-6 centered rectangle shifted right by 7.

Answer: rectangle formula plus time shift

For the same rectangle centered at zero, $R(j\omega) = \dfrac{2\sin(3\omega)}{\omega}$.

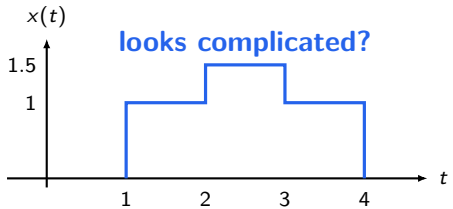
Our signal, $x(t) = r(t - 7)$.

Using time shifting, $r(t - t_0) \longleftrightarrow e^{-j\omega t_0} R(j\omega)$.

$$X(j\omega) = e^{-j7\omega} \frac{2\sin(3\omega)}{\omega}$$

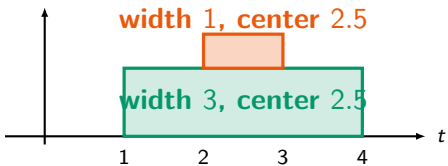
A Lego block signal

$$x(t) = \begin{cases} 1, & 1 < t < 2, \\ 1.5, & 2 < t < 3, \\ 1, & 3 < t < 4, \\ 0, & \text{otherwise.} \end{cases}$$



Decompose the Lego block

$$x(t) = \underbrace{\text{height 1 rectangle from 1 to 4}}_{\text{wide block}} + \underbrace{\text{height 0.5 rectangle from 2 to 3}}_{\text{small top block}}.$$



Fourier transform of the Lego block

Wide block: width 3, center 2.5 $\Longrightarrow$ $e^{-j2.5\omega}\dfrac{2\sin(1.5\omega)}{\omega}$.

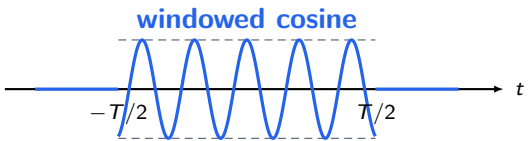
Top block: height 0.5, width 1, center 2.5 $\Longrightarrow$ $0.5e^{-j2.5\omega}\dfrac{2\sin(0.5\omega)}{\omega}$.

$$X(j\omega) = e^{-j2.5\omega}\left[\frac{2\sin(1.5\omega)}{\omega} + \frac{\sin(0.5\omega)}{\omega}\right]$$

Now lock a cosine inside the rectangle

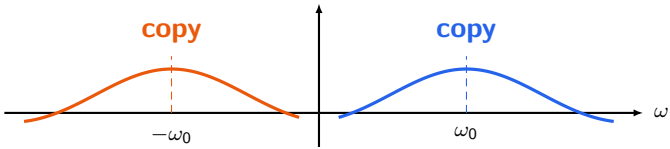
Let $r(t)$ be the rectangle from $-T/2$ to $T/2$.

$$x(t) = r(t)\cos(\omega_0 t)$$



This is just modulation of a rectangle.

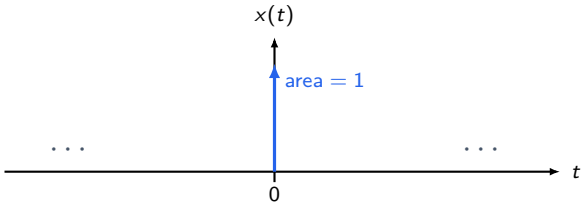
A windowed cosine creates two sinc-shaped copies



A cosine inside a rectangle has spectrum centered around $\pm\omega_0$, not around zero.

Another example: Fourier transform of a spike

$$x(t) = \delta(t)$$



What does this single spike look like in the frequency domain?

Simple, let's use the sifting property!

$$\text{For } x(t) \quad = \quad \delta(t),$$

$$X(j\omega) = \int_{-\infty}^{\infty} \delta(t) e^{-j\omega t}\, dt.$$

Now the impulse simply samples the background function at $t = 0$:

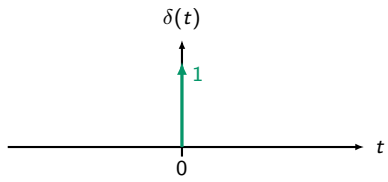
$$\int_{-\infty}^{\infty} \delta(t) f(t)\, dt = f(0).$$

$$X(j\omega) \quad = \quad e^{-j\omega \cdot 0} \quad = \quad 1.$$

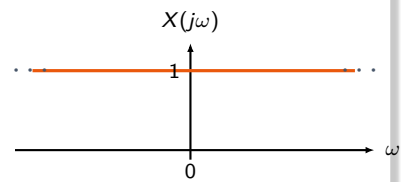
So the spectrum is completely flat

$$\delta(t) \quad \longleftrightarrow \quad 1$$

Time domain



Frequency domain



A very narrow spike in time needs all frequencies equally.

Small bonus: what if the spike is shifted?

Let, $x(t) = \delta(t - t_0)$.

Then, $X(j\omega) = \int_{-\infty}^{\infty} \delta(t - t_0) e^{-j\omega t}\, dt$.

Using sifting, the impulse samples the exponential at $t = t_0$:

$$X(j\omega) = e^{-j\omega t_0}.$$

$$\boxed{\delta(t - t_0) \quad \longleftrightarrow \quad e^{-j\omega t_0}}$$

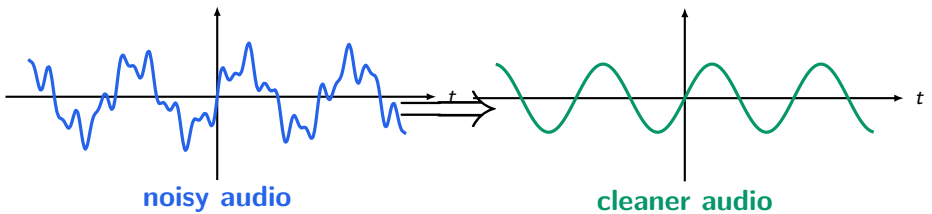
Applications

Where Fourier transform quietly appears

Audio Denoising

Removing the unwanted frequencies

Audio denoising: keep the voice, reduce the noise



Identify and suppress unwanted high-frequency noise in the Fourier spectrum, then apply the inverse Fourier transform to reconstruct cleaner audio.

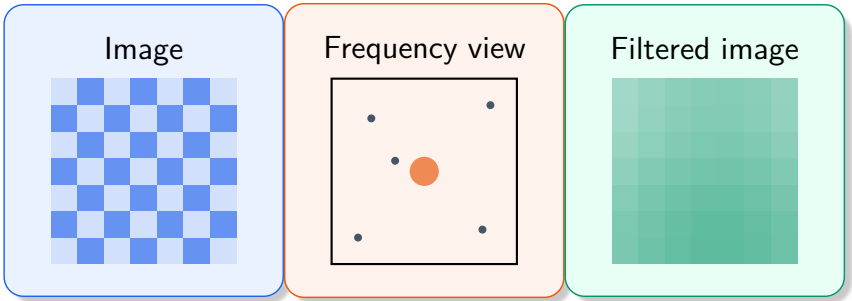
Please try this visualization on Audio Denoising!



Image Denoising

Clean an image by editing its frequencies

Image denoising: noise often hides in high frequencies



Suppress unwanted high-frequency components, then apply the inverse Fourier transform to reduce pixel-level noise while preserving as much image detail as possible.

Please try this visualization on Image Denoising!

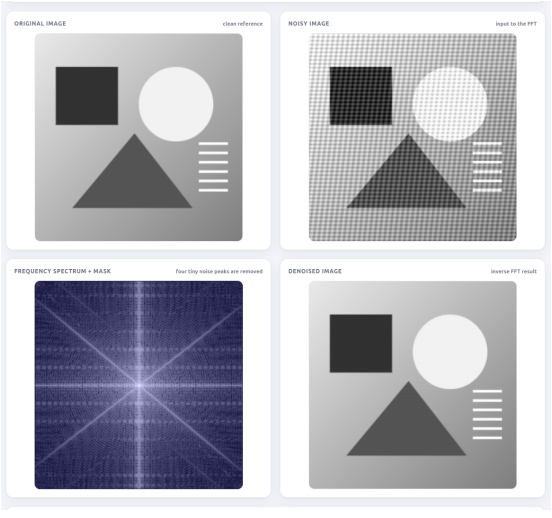
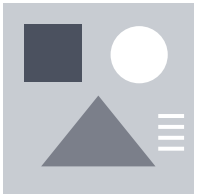


Image Edge Detection

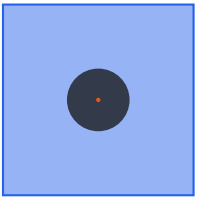
Finding boundaries by keeping rapid changes

Edge detection: boundaries hide in high frequencies

Original image



High-pass filter



Detected edges



Suppress low spatial frequencies, then apply the inverse Fourier transform.

Please try this visualization on Image Edge Detection!

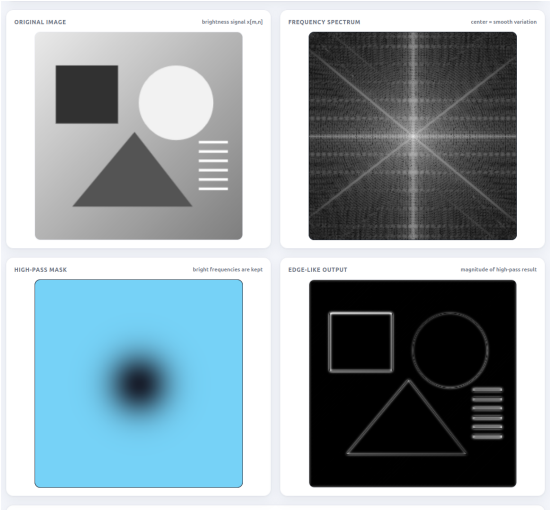
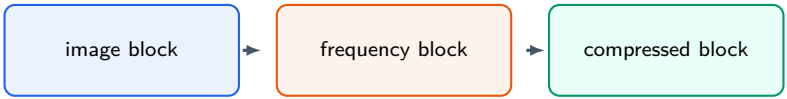


Image and Audio Compression

Let's throw away what the eye or ear can barely notice

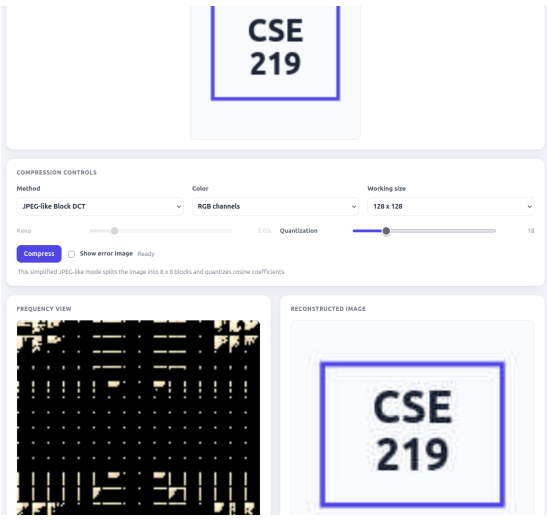
Compression: not all frequencies are equally important



small high-frequency coefficients are rounded away

Transform images or audio into frequency components, discard or approximate less important coefficients, and reconstruct the signal using fewer bits while preserving perceptual quality.

Please try this visualization on Image Compression!



Thank you!

Thank youuu! Until we meet again...

